

Uttam Mahata

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EDUCATION

Indian Institute of Engineering Science & Technology, Shibpur

Howrah, India

Bachelor of Technology – Computer Science and Technology

Aug. 2022 – May 2026

- **CGPA: 9.07 / 10.0** (Till 7th Semester)
- Relevant Coursework: Data Structures & Algorithms, Computer Architecture & Organization, Operating Systems, Compiler Design, Computer Networks, Database Management Systems, Machine Learning, Software Engineering, Information Security & Cryptography, Quantum Computing

PROFESSIONAL EXPERIENCE

OpenLM

Kolkata, India

Engineer Trainee

Jan. 2026 – Present

- Tuned **SparkApplication Custom Resource Definition (CRD)** executor memory and overhead allocations, eliminating frequent Out-of-Memory (OOM) kills and improving pipeline stability by **90%+** under high-workload conditions; diagnosed and fixed a **PermissionDenied** job failure caused by non-root container path ownership gaps in the Dockerfile.
- Authored **4 end-to-end Azure DevOps CI/CD pipelines** covering a schema-migration Kubernetes Job and a multi-tenant **Kubernetes CronJob** for Spark load testing on staging environments.
- Rightsized **Kubernetes** resource requests and limits across **30+ microservices** using Datadog CPU/memory dashboards, resolving recurring OOMKills and stabilising workload performance.
- Developed an autonomous incident-response system using **Google Agent Development Kit (ADK)** and **Model Context Protocol (MCP)** – an AI orchestrator agent inspects logs and downstream service health via Datadog, performs root cause analysis, and auto-creates tickets in **Azure DevOps**, acting as an intelligent on-call Site Reliability Engineer (SRE) powered by a **Large Language Model (LLM)**.

Alpha AI Service Pvt Ltd

New Delhi, India (Remote)

Software & DevOps Intern

Jun. 2025 – Aug. 2025

- Implemented a **Factory Provider** design pattern to dynamically adapt AI providers (**Gemini API**, **LangChain**) at runtime, supporting image, text-to-speech (TTS), and video content generation across 10+ content types on a production platform.
- Delivered a **credit-based / usage-based billing system** integrated with **Razorpay**, metering per-generation consumption and automating payment collection across subscription tiers.
- Deployed and maintained a highly-available **K3s** cluster across 4 VPS nodes (2 control planes, external MySQL datastore, AES-CBC secret encryption) with all inter-node traffic over a **WireGuard** full-mesh VPN.
- Provisioned CI/CD with **8 self-hosted GitHub Actions runners** and a private Docker registry with daily GC; authored a **GitOps** deployment engine – 1 Ansible role + 10 Jinja2 templates deploying **23 microservices** across 3+ products with **Infisical**-synced secrets and annotation-driven zero-touch rolling updates.

PROJECTS

DilithiAuth – Post-Quantum OAuth 2.0 Identity Provider | Go, ML-DSA-44, Ed25519, Redis,

Mar. 2026 – Present

PostgreSQL, React, Docker

- Architected a post-quantum OAuth 2.0 / OpenID Connect (OIDC) identity provider that dual-signs every token with **Ed25519 + ML-DSA-44 (NIST FIPS 204)**, protecting against Harvest-Now-Decrypt-Later attacks where encrypted data is collected today and decrypted using future quantum computers.
- Engineered three token formats to manage large post-quantum signature size (approx. 2.4 KB): a compact Thin JSON Web Token (JWT, approx. 500B) with server-side signature storage, a self-contained Jumbo JWT (3–4 KB), and a binary **CBOR Web Token** (1.2 KB) – each selectable based on infrastructure constraints.
- Implemented an **Edge Proxy** gateway that handles all post-quantum cryptography (PQC) verification centrally, strips the large signature, and forwards a standard lightweight token – keeping all internal microservices fully PQC-agnostic with no cryptographic changes required.

RootAccess – CTF Competition Platform | Go, Angular 21, MongoDB, Redis, AWS Lambda,

May 2025 – Present

OCI, GCP, Cloudflare

- Architected and shipped ctf.rootaccess.live, a full-stack Capture The Flag platform using **Go (Gin)** API and **Angular 21** frontend with real-time scoreboards via **WebSocket** and **Upstash Redis** pub/sub.
- Implemented dynamic scoring (CTFd formula), **JWT** auth with HTTP-only cookies, per-IP rate limiting, email verification, and **role-based access control (RBAC)** for multi-team competition.
- Deployed multi-cloud infrastructure across **Google Cloud Functions** and **AWS Lambda** behind **Cloudflare Workers**, delivering 3M free requests/month; orchestrated quota failover with **GitHub Actions** daily cron.

CourseWagon – Generative AI Course Authoring & LMS | *Angular, FastAPI, Gemini API, Firebase, Docker, GCP* *Aug. 2024 – Present*

- Shipped an enterprise-grade LMS that auto-generates full course hierarchies (subjects, chapters, and topics) using **Gemini API**, cutting manual curriculum-authoring effort by **60%**.
- Integrated **Firebase Auth + JWT**, **Oracle Cloud Object Storage** for media, KaTeX/MathJax for equations, and Mermaid.js for diagrams; deployed on **GCP** with **Google Cloud Run**.

Winners Education – Coaching Institute Management System | *Angular, Spring Boot, MySQL* *Jun. 2023 – Jun. 2024*

- Built a full-stack institute management system using **Angular**, **Spring Boot**, and **MySQL**, enabling course and exam tracking for **100+ students** with secure role-based admin and student modules.
- Developed a scoreboard management system with **PDF export** for question papers and result reports, leveraging Spring Boot PDF libraries to eliminate manual offline processing.

SKILLS

Programming Languages: C, Java, Python, TypeScript, Go (Golang)

Frontend: Angular, React, HTML, CSS, JavaScript

Backend: Spring Boot, FastAPI, Gin (Go), Flask, NestJS

Databases: MySQL, PostgreSQL, MongoDB, Firebase Firestore, Redis, SQLite

Cloud & DevOps: Linux, Docker, Kubernetes, EKS, K3s, Helm, Ansible, Apache Spark, WireGuard, Traefik, GCP, AWS, Azure DevOps, Cloudflare, GitHub Actions, CI/CD

Technical Skills: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Computer Architecture & Organization, Cryptography

Others: Google Agent Development Kit (ADK), LangChain, LangGraph, Large Language Model (LLM) Integration, Retrieval-Augmented Generation (RAG)

ACHIEVEMENTS

Brain Dead Hackathon – IEST Shibpur

Howrah, India

2nd Place out of 123 teams – Inter-University Data Science & Machine Learning Hackathon

Mar. 2025

IEST-UCO Bank Hackathon

Howrah, India

Designed and built Aegis, a production-grade secure-banking prototype sponsored by UCO Bank

Jul. 2025 – Aug. 2025

EOFool CTF – Instruo 2025, Inter-University Tech Fest

Howrah, India

3rd Place out of 40 teams – Capture The Flag competition

Sep. 2025